Exp No:4 Experiment to understand the data preprocessing in Data science

CODE

```
import numpy as np

import pandas as pd

df = pd.DataFrame({

   'Country':
['India','Morocco','Germany','Morocco','Germany','India','Morocco','India','Germany','India'],

   'Age': [44, 27, 30, 38, 40, 35, np.nan, 48, 50, 37],

   'Salary': [72000, 48000, 54000, 61000, np.nan, 58000, 52000, 79000, 83000, 67000],

   'Purchased': ['No', 'Yes', 'No', 'No', 'Yes', 'Yes',

            'No', 'Yes', 'No', 'Yes']

})

print("Original Dataset:")

print(df)

print("\n\nDataFrame Information:")

print(df.info())

print("\n\nCountry Mode:")

print(df.Country.mode())

print("\n\nFirst Mode Value:")

print(df.Country.mode()[0])

df['Country'] = df['Country'].fillna(df.Country.mode()[0])
```

```
df['Age'] = df['Age'].fillna(df.Age.median())
df['Salary'] = df['Salary'].fillna(round(df.Salary.mean()))
print("\n\nDataset After Filling Missing Values:")
print(df)
print("\n\nOne Hot Encoding:")
print(pd.get_dummies(df.Country))
updated_dataset = pd.concat(
   [pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
   axis=1
)
print("\n\nUpdated Dataset:")
print(updated_dataset)
print("\n\nDataFrame Information After Preprocessing:")
print(df.info())
updated_dataset['Purchased'] =
updated_dataset['Purchased'].replace(
   ['No', 'Yes'], [0, 1]
)
print("\n\nFinal Dataset:")
print(updated_dataset)
```

OUTPUT

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    9 non-null      object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	France	50.0	83000.0	No

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	True	False	False	50.0	83000.0	No

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0

RESULT hence data preprocessing in Data science has been studied successfully.